SG No. 113 D_{2d}^3 $P\bar{4}2_1m$ [tetragonal]

* plus set: $+[0, 0, 0]$

Table 1: Wyckoff site: `2a`, site symmetry: `-4..`

No.	position	mapping
1	$[0, 0, 0]$	`[1,2,3,4]`
2	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[5,6,7,8]`

Table 2: Wyckoff site: `2b`, site symmetry: `-4..`

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	`[1,2,3,4]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[5,6,7,8]`

Table 3: Wyckoff site: `2c`, site symmetry: `2.mm`

No.	position	mapping
1	$\left[0, \frac{1}{2}, z\right]$	`[1,2,7,8]`
2	$\left[\frac{1}{2}, 0, -z\right]$	`[3,4,5,6]`

Table 4: Wyckoff site: `4d`, site symmetry: `2..`

No.	position	mapping
1	$[0, 0, z]$	`[1,2]`
2	$[0, 0, -z]$	`[3,4]`
3	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	`[5,6]`
4	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[7,8]`

Table 5: Wyckoff site: `4e`, site symmetry: `..m`

No.	position	mapping
1	$\left[x, x + \frac{1}{2}, z\right]$	`[1,8]`
2	$\left[-x, \frac{1}{2} - x, z\right]$	`[2,7]`
3	$\left[x + \frac{1}{2}, -x, -z\right]$	`[3,6]`
4	$\left[\frac{1}{2} - x, x, -z\right]$	`[4,5]`

Table 6: Wyckoff site: `8f`, site symmetry: `1`

No.	position	mapping
1	$[x, y, z]$	`[1]`
2	$[-x, -y, z]$	`[2]`
3	$[y, -x, -z]$	`[3]`
4	$[-y, x, -z]$	`[4]`

continued ...

Table 6

No.	position	mapping
5	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, -z\right]$	[5]
6	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, -z\right]$	[6]
7	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, z\right]$	[7]
8	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, z\right]$	[8]